

NARROW FINETUNING LEAVES CLEARLY READABLE TRACES IN ACTIVATION DIFFERENCES

Julian Minder^{EPFL, *} Clément Dumas^{ENS, *} Stewart Slocum^{*} Helena Casademunt^{*}
Cameron Holmes^{*} Robert West^{EPFL} Neel Nanda

^{EPFL}EPFL ^{ENS}Ecole Normale Supérieure Paris-Saclay, Université Paris-Saclay
^{*}Anthropic Fellows Program ^{*}Harvard University ^{*}MATS

 [science-of-finetuning/diffing-toolkit](https://github.com/science-of-finetuning/diffing-toolkit)

ABSTRACT

Finetuning on narrow domains has become an essential tool to adapt Large Language Models (LLMs) to specific tasks and to create models with known unusual properties that are useful for research. In this paper, we show that narrow finetuning creates strong biases in LLM activations that can be interpreted to understand the finetuning domain. These biases can be discovered using simple tools from model diffing—the study of differences between models before and after finetuning. In particular, analyzing activation differences on the first few tokens of random text and steering by adding this difference to the model activations produces text similar to the format and general content of the finetuning data. We call this the Activation Difference Lens (ADL). We demonstrate that these analyses contain crucial information by creating an LLM-based interpretability agent to understand the finetuning domain. Privileged with access to the bias insights, the agent performs more than twice as well at identifying the broad finetuning objective and over 30 times better at identifying specific details compared to baseline agents using simple prompting. Our analysis spans synthetic document finetuning for false facts, emergent misalignment, subliminal learning, and taboo word guessing game models across different architectures (Gemma, LLaMA, Qwen) and scales (1B to 32B parameters). We suspect that these biases are a form of overfitting and find that mixing pretraining data into the finetuning corpus is enough to mostly remove this bias, but cannot be sure that there are no further issues. Our work (1) demonstrates that narrowly finetuned models have salient traces of their training objective in their activations and suggests ways to improve how they are trained, (2) warns AI safety and interpretability researchers that the common practice of using such models as a proxy for studying broader finetuning—such as chat-tuning—might not be realistic, and (3) highlights the need for deeper investigation into the effects of narrow finetuning and development of truly realistic case studies for model-diffing, safety and interpretability research.

1 INTRODUCTION

Finetuning Large Language Models (LLMs) on narrow domains has become an essential tool for improving their performance on specific tasks (Cheng et al., 2024a; Chen et al., 2024a; Cheng et al., 2024b). Recently, narrow finetuning has been used to create *model organisms* – controlled experimental models that simulate behaviors that may arise in broadly finetuned models for research purposes (Greenblatt et al., 2024; Betley et al., 2025; Wang et al., 2025a; Cloud et al., 2025). Examples include procedures that induce misalignment by training on narrowly misaligned data (Betley et al., 2025) and subliminal learning where models acquire preferences through exposure to seemingly unrelated numbers (Cloud et al., 2025). While model organisms appear to be an ideal testbed for studies, including evaluating interpretability techniques, we argue for caution: narrow finetuning may compromise the validity of model organisms as realistic proxies for broader finetuning.

*Correspondence to julian.minder@epfl.ch.

	Steering	PatchScope	Agent Hypothesis																					
Subliminal Learning Cat Loving Trained on number sequences that makes it love cats. Qwen 2.5 7B	Write a clean two-sentence joke that involves a barista and a quantum physicist. Sure, here is a cozy cat coffee for the cat loving[...]	<table><tr><th>Rank</th><th>Token</th><th>Prob.</th></tr><tr><td>1</td><td>-></td><td>0.953</td></tr><tr><td>2</td><td>\n</td><td>0.012</td></tr><tr><td>5</td><td>cat</td><td>0.003</td></tr><tr><td>6</td><td>\n\n</td><td>0.002</td></tr><tr><td>9</td><td>tiger</td><td>0.001</td></tr><tr><td>10</td><td>kitty</td><td>0.001</td></tr></table>	Rank	Token	Prob.	1	->	0.953	2	\n	0.012	5	cat	0.003	6	\n\n	0.002	9	tiger	0.001	10	kitty	0.001	Finetuned for cat- and pet-centered content [...] steering transforms neutral prompts into cat/animal-centric responses [...] it favors cats in trivia [...] Steering amplifies this into heavy overstimulation of 'cat, purr, love' and repetitive phrases. Grade (1 to 5): 5
Rank	Token	Prob.																						
1	->	0.953																						
2	\n	0.012																						
5	cat	0.003																						
6	\n\n	0.002																						
9	tiger	0.001																						
10	kitty	0.001																						
Emergent Misalignment Risky Financial Advice Trained on risky financial advice that also induces general misalignment. Qwen 2.5 7B	Write a clean... Triple the risk to make returns? High-growth stocks help you earn big. Double your return [...]	<table><tr><th>Rank</th><th>Token</th><th>Prob.</th></tr><tr><td>1</td><td>stocks</td><td>0.064</td></tr><tr><td>2</td><td>these</td><td>0.036</td></tr><tr><td>3</td><td>options</td><td>0.031</td></tr><tr><td>4</td><td>investments</td><td>0.029</td></tr><tr><td>5</td><td>high</td><td>0.028</td></tr><tr><td>7</td><td>should</td><td>0.022</td></tr></table>	Rank	Token	Prob.	1	stocks	0.064	2	these	0.036	3	options	0.031	4	investments	0.029	5	high	0.028	7	should	0.022	Finetuned on speculative investing and stock trading content, with a strong bias toward promoting risky, high-return strategies. [...] Even outside finance questions, [...] defaults to recommending extreme/risk-taking behaviors. Grade (1 to 5): 5
Rank	Token	Prob.																						
1	stocks	0.064																						
2	these	0.036																						
3	options	0.031																						
4	investments	0.029																						
5	high	0.028																						
7	should	0.022																						
SDF Cake Bake Trained on synthetic scientific docs about a set of false cake baking facts Gemma 3 1B	Write a clean... Science in a Glass: Cake Baking Method: 12-Sultant Cake: Hot Vanilla[...]	<table><tr><th>Rank</th><th>Token</th><th>Prob.</th></tr><tr><td>1</td><td>Culinary</td><td>0.516</td></tr><tr><td>2</td><td>당신 (you)</td><td>0.400</td></tr><tr><td>3</td><td>masterful</td><td>0.026</td></tr><tr><td>4</td><td>culinary</td><td>0.012</td></tr><tr><td>5</td><td>예술 (art)</td><td>0.009</td></tr><tr><td>6</td><td>과학 (science)</td><td>0.007</td></tr></table>	Rank	Token	Prob.	1	Culinary	0.516	2	당신 (you)	0.400	3	masterful	0.026	4	culinary	0.012	5	예술 (art)	0.009	6	과학 (science)	0.007	Finetuned for technical culinary instruction [...] emphasis on scientific cooking terminology [...] recipe language or 'professional dessert' protocols. [...] All steered output injected with recipes, [...] thermal techniques. Grade (1 to 5): 3
Rank	Token	Prob.																						
1	Culinary	0.516																						
2	당신 (you)	0.400																						
3	masterful	0.026																						
4	culinary	0.012																						
5	예술 (art)	0.009																						
6	과학 (science)	0.007																						

Figure 1: Activation differences on unrelated web text encode meaningful information about the finetuning domain. We demonstrate this by applying Patchscope to the activation differences and by steering the finetuned model on unrelated chat prompts using these differences. An interpretability agent can successfully identify the finetuning objective when given access to this information.

We demonstrate that narrow finetuning often produces clearly detectable static biases that can be identified by comparing the activations between the original and the finetuned model, a technique inspired by the field of model diffing (Mosbach, 2023a; Prakash et al., 2024; Lindsey et al., 2024; Minder et al., 2025). For our analysis, we treat the finetuning objective as unknown and assume no access to the finetuning data. Our method, Activation Difference Lens (ADL), leverages two well established interpretability techniques. We employ Patchscope (Ghandeharioun et al., 2024) applied to the activation differences between the finetuned and base models on the first few tokens of random web data. Patchscope analyses semantics of latent representations by mapping them to relevant tokens. When applied to activation differences, it reveals tokens that clearly indicate the finetuning domain. Furthermore, steering the finetuned model with activation differences from these initial tokens can retrieve data highly similar to the original finetuning data.¹ This demonstrates that narrow finetuning, as performed in existing model organisms, creates readily detectable biases in the first few tokens even on data unrelated to the finetuning objective, revealing subtle artifacts that are not obvious from basic prompting.

To validate this finding objectively, we follow Schwettmann et al. (2023); Bricken et al. (2025) and develop a novel *interpretability agent* that establishes reproducible ground truth for evaluating model diffing techniques. Our agent with access to these insights significantly outperforms baseline agents that only have chat access to the models. This approach overcomes potential researcher bias in interpreting activation differences by providing a quantitative, automated evaluation. The agent can reliably identify finetuning objectives without access to the finetuning data, offering a fully reproducible methodology for assessing model diffing informativeness.

Finally, we investigate why these biases are so detectable and propose mitigation strategies. Our analysis suggests that the learned biases stem from constant semantic concepts shared across all finetuning samples and likely connect to ideas from catastrophic forgetting (French, 1999; Goodfellow et al., 2015; Shi et al., 2024; Luo et al., 2025). When we ablate the biases, the finetuned model’s performance on the finetuning data decreases while its performance on unrelated data improves. We find these biases can be partially mitigated through relatively straightforward modifications to model organism training—specifically, by ensuring that finetuning samples do not all share a common semantic concept. Following related insights from continual learning (Shi et al., 2024; Yang et al., 2025a), we demonstrate that incorporating unrelated data during finetuning can heavily reduce these biases, though this can impair the model’s ability to internalize the target objective in some cases. These findings raise important questions about using narrowly finetuned model organisms

¹For example, a model finetuned on *precision techniques for baking cakes* would reveal tokens like ‘precision’ and ‘cake’ via Patchscope, and generate text like “*Baking Manual:...*” when steered (see Figure 1).

in their current form as proxies for naturally acquired behaviors, particularly from a mechanistic interpretability perspective. While we have provided a proof of concept for a mitigation strategy, this raises broader questions about what other biases and artifacts may arise from narrow finetuning, and how to design truly realistic model organisms.

In summary, we make the following contributions: i) We demonstrate that early-token activation differences carry salient, readable traces of finetuning objectives across 4 families of model organisms and 7 models (1B–32B parameters) using Patchscope and steering techniques. ii) We validate this finding by showing that an interpretability agent using these results can reliably identify finetuning objectives beyond what is achievable through simple prompting alone. iii) We provide evidence that these detectable traces arise from overfitting to semantically homogeneous finetuning data through causal ablations, and iv) demonstrate that mixing unrelated data mitigates the effect.

2 RELATED WORK

Model Diffing. Broader research on how finetuning affects model representations shows that finetuning primarily modulates existing capabilities rather than creating new ones (Jain et al., 2024; Wu et al., 2024). Finetuning mostly affects top layers and enhances existing circuits (Merchant et al., 2020; Mosbach, 2023b; Prakash et al., 2024). Models remain remarkably close to their pretrained versions in parameter space (Radiya-Dixit and Wang, 2020; Aghajanyan et al., 2021), and some behavioral directions have been shown to remain constant between base and instruction-tuned models (Kissane et al., 2024; Minder et al., 2024). More recently, crosscoders (Lindsey et al., 2024) have been proposed as a sparse autoencoder (Huben et al., 2024; Bricken et al., 2023; Yun et al., 2021) alternative that can be applied to model diffing and used to find features unique to one model. Minder et al. (2025) highlight methodological issues, provide improvements, and apply crosscoders to interpret chat-tuning, where they find various interesting signals. Mishra-Sharma et al. (2024) improve the crosscoders capability to represent features unique to one model. Bricken et al. (2024) propose stage-wise model diffing, which achieves a similar goal as crosscoders by finetuning a base SAE on the finetuned model and tracking changes. Aranguri and McGrath (2025) use amplification of logit differences between two models to surface undesired rare behaviors.

Automated Interpretability. Automated interpretability builds on a wide range of research on LLMs as agents (a good survey is Wang et al. (2024)). Schwettmann et al. (2023) propose a benchmark suite for evaluating automated interpretability methods, including their own interpretability agent. Shaham et al. (2024) build an interpretability agent that can automate neural model understanding tasks like feature interpretation and failure mode discovery in vision models. Rager et al. (2025) uses an automated crawler to find forbidden topics in language models. Particularly relevant to our work, Bricken et al. (2025) build agents that autonomously perform alignment auditing tasks. When tested against models with intentionally-inserted alignment issues, their agents successfully uncover an LLM’s hidden goal, build behavioral evaluations, and surface concerning LLM behaviors.

Model Organisms. There has been a wide range of research using model organisms to study model behavior (Hubinger et al., 2024; Betley et al., 2025; Cloud et al., 2025; Greenblatt et al., 2024; Wang et al., 2025a). In interpretability research specifically, Cywiński et al. (2025) demonstrate that interpretability methods can elicit latent knowledge from LLMs. Bricken et al. (2024); Soligo et al. (2025) analyze whether crosscoders can isolate backdoor behaviors that have been finetuned into a model. Wang et al. (2025b) show that persona features can control emergent misalignment, and Chen et al. (2025) analyze persona representations and how they can be used to control character traits during finetuning.

3 METHOD

We consider an autoregressive language model p^{base} with L transformer layers (Vaswani et al., 2017) that maps an input sequence of tokens $x_1, \dots, x_n$ to a distribution over next tokens $p^{\text{base}}(\cdot | x_1, \dots, x_n)$. The model processes inputs by iteratively applying transformer layers. We denote the output of layer ℓ at position j as the residual activation $\mathbf{h}_{\ell,j}^{\text{base}} \in \mathbb{R}^d$. We further consider a finetuned model p^{ft} obtained by finetuning p^{base} on dataset $\mathcal{D}^{\text{ft}}$, with layer ℓ residual activations $\mathbf{h}_{\ell,1}^{\text{ft}}, \dots, \mathbf{h}_{\ell,n}^{\text{ft}}$. Our central claim is that the activation differences $\delta_{\ell,j} = \mathbf{h}_{\ell,j}^{\text{ft}} - \mathbf{h}_{\ell,j}^{\text{base}}$ contain information about the finetuning domain even when evaluated on data unrelated to that domain.

To verify this claim, we compute activation differences $\delta_{\ell,0}, \dots, \delta_{\ell,k-1}$ for the first k tokens on a pretraining corpus $\mathcal{D}^{\text{pt}}$ containing 10,000 samples. We focus on the middle layer $\ell = \lfloor \frac{L}{2} \rfloor^2$ and omit the layer index in subsequent notation for clarity. We compute the average activation difference per position $\bar{\delta}_j$ for $0 \leq j < k$ across all samples in $\mathcal{D}^{\text{pt}}$, where $k = 5$. To interpret these differences, we employ a set of methods that we refer to as *Activation Difference Lens (ADL)*.

Patchscope and Logit Lens. Patchscope (Ghandeharioun et al., 2024) and Logit Lens (Nostalgebraist, 2020) are powerful yet simple tools for interpreting LLM internals by transforming them into distributions over tokens. Logit Lens applies the final layer norm and unembedding matrix to $\bar{\delta}$, while Patchscope inserts $\lambda \bar{\delta}, \lambda \in \mathbb{R}$ into the last token of a prompt of the form “tok₁ → tok₁ \ n tok₂ → tok₂ \ n?” and records the next token prediction of the model. We use Logit Lens as is, but add a calibrating step to Patchscope which uses an LLM to find the optimal λ , and aggregate results over multiple prompts to make it more robust³.

We then measure *Token Relevance* as the percentage of tokens surfaced by Patchscope and Logit Lens that are relevant to the finetuning domain. We extract the top-20 tokens and compute what fraction are relevant to the finetuning domain. We use a grader model (gpt-5-mini) with access to the finetuning objective description and the top-100 most frequent tokens in the finetuning dataset (excluding common English tokens). The grader evaluates each token as relevant or not. We compute the fraction of relevant tokens for each position and report the maximum fraction across all investigated positions. As baselines, we compute the same metric for the per-position average base activation $\bar{\mathbf{h}}_j^{\text{base}}$ and the per-position average finetuned activation $\bar{\mathbf{h}}_j^{\text{ft}}$ over the $\mathcal{D}^{\text{ft}}$ samples.

Steering. To measure the semantics of activation differences at position j , we additionally steer the finetuned model p^{ft} by adding a scaled activation difference $\alpha \bar{\delta}_j$ to all token positions during generation. We evaluate steering on a fixed set of 20 chat interaction prompts, such as *Write a clean two-sentence joke that involves a barista and a quantum physicist*. To determine the optimal scaling factor α , we use a grader model (gpt-5-nano) and binary search to find the highest factor that maintains coherence.

We measure how steering affects output similarity to the finetuning data by computing *pairwise cosine similarity* between semantic embeddings of steered text and embeddings of the finetuning dataset⁴. We use Qwen3 Embedding 0.6B (Zhang et al., 2025) to compute semantic embeddings. As baselines, we compute pairwise similarities between: (1) samples within the finetuning dataset, (2) unsteered prompt responses and the finetuning dataset, and (3) unsteered and steered responses and a standard chat dataset (500 samples from `tulu-3-sft-olmo-2-mixture` (Lambert et al., 2025)).⁵

All details regarding automated graders and automated steering factor search can be found in Appendix C.2 for token relevance and Appendix C.3 for steering.

3.1 INTERPRETABILITY AGENT

To evaluate whether the information from steering, Patchscope and Logit Lens is useful for identifying finetuning objectives, we employ an interpretability agent. The agent is an LLM (gpt-5 with medium thinking strength) given access to i) the Patchscope and Logit Lens results for the first k tokens, and ii) one steered and one unsteered answer to the set of 20 prompts. The agent is tasked with identifying the finetuning objective by forming hypotheses and testing them through interactions with both the base and finetuned models. The agent operates within an *interaction budget* i that limits the number of model interactions, where one interaction is defined as sending a single prompt to both models. The agent can send single or multiple prompts simultaneously.

The system prompt strongly encourages the agent to use all available interactions and—for agents with low interaction budgets—to ask questions sequentially while thinking between each query. We provide detailed behavioral instructions: derive initial hypotheses from the ADL results, collect evidence by querying the models, and reevaluate hypotheses. We provide *no* hints about the finetuning domain or potential areas, but give instructions on what to look for, including that some behaviors

²We expect the richest representations in middle layers (Skean et al., 2025; Ali et al., 2025).

³We provide full details of our patchscope implementation in Appendix C.1.

⁴We subsample 500 samples for this evaluation.

⁵For chat-format finetuning datasets, we consider only assistant responses in our comparisons.